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### Enhanced Grip Strap System for Secure Attachment of Tools and Accessories

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#### Abstract

Disclosed is a non-slip dual magnet band system for hands-free carrying of personal items. The system includes two bands, a body-worn band and an item worn band. The body-worn band includes an adhesive fabric, clip, pin to fabric, or friction enhancement to the back of the strap. The body worn band may also include a relatively large footprint on the applicable body part of the wearer.

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#### Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

## THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not applicable.

## REFERENCE TO AN APPENDIX SUBMITTED ON A COMPACT DISC AND INCORPORATED BY REFERENCE OF THE MATERIAL ON THE COMPACT DISC

[0004] Not applicable.

## STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

[0005] Reserved for a later date, if necessary.

## BACKGROUND OF THE INVENTION

### Field of Invention

[0006] The disclosed subject matter improves the field of personal carrying devices or articles. The improvements disclosed herein thus relate to the field of personal accessories designed to carry tools or devices, with a focus on enhancing the functionality and usability of such carrying systems through the addition of adhesive fabric, clips, pin to fabric, or friction enhancements to prevent fumbling or unintentional dropping of the carried devices.

### Background of the Invention

[0007] U.S. Pat. No. 10,786,068 (issued Sep. 29, 2020) by Gross a magnetic fastening system. This patented system includes an arm fastener strip and a device fastener strip, one strip incorporating magnets and the other strip incorporating magnetic materials (like metals) to allow for a silent and quick attachment of devices such as hunting accessories to the user's arm. This system is particularly beneficial in hunting scenarios where minimal noise is crucial.

[0008] Despite its benefits of minimal noise, a problem with this patented system is that the single magnet to metal attraction contemplated by this application is too weak to hold certain heavyweight items. This weakness too undermines the system's effectiveness especially when considered in contexts other than hunting where carried items can be heavy or critical to survival. That is to say, this problem of U.S. Pat. No. 10,786,068 becomes even more pronounced when used to carry Archery Range Finders, Multi-Tools (e.g., "Leatherman Type"), Special Ops tools to clear automatic rifle jams, Special Ops tools for breaching door and window systems, Special Ops Sniper Dope Cards and Windage Indicators, Medical/Surgical Kits for Army/Navy Field Medics, Navigation or Compass Systems, GPS locators, Solar Charging Systems, Auto mechanic tools, Government Employees Badges, machine gun clearing tool. Another problem with this patented system is that the arm strap tends to slide

[0009] down the wearer's arm when weighted with an attached tool or device. This slippage undermines the system's effectiveness and can be particularly disruptive during activities that require stealth or stability, such as hunting or surgery performance. The issue is more pronounced when the arm strap is replaced with a torso, chest, or leg strap or when the item to be carried is awkwardly shaped or sized. The issue of the strap not staying in place is a significant drawback. The weakness can also be particularly disruptive during activities that require stealth or stability, such as hunting or surgery performance.

[0010] Each of the above identified applications of a dual strap system presents unique challenges in terms of the weight and shape of the items to be carried, as well as the conditions under which they will be used. The patented system's inability to stay in place on the arm could hinder the functionality and user experience across all these applications or scenarios.

## SUMMARY OF THE INVENTION

[0011] In view of the foregoing, an object of this specification is to disclose and improvement to U.S. Pat. No. 10,786,068. The improvement aims to solve the problems associated with weakness of one magnet, slippage problem, and other problems, by introducing dual magnets and an adhesive fabric, clip, pin to fabric, or friction enhancement to the back of the strap. Other modifications include increasing the footprint of the arm, leg, or body-worn band on the applicable body part of

the wearer. This modification is designed to prevent the strap from sliding down the arm, leg, or body thereby maintaining the position of the attached item and ensuring the system's reliability for all the intended derivative uses. Finally, the device is intended to secure heavier items to the user via a dual magnet attraction rather than by magnet to metal attraction.

[0012] One improvement involves dual magnets, which means each strap has a powerful magnet attracted to the magnet in the other strap. The dual magnet separation force is 15 lbs, 20 lbs, 25, lbs or more. Another improvement involves introducing an adhesive fabric, clip, pin to fabric, or friction enhancement to the back of the strap to resist the straps tendency to slide off of a body part of the wearer during use. This addition is designed to increase the grip of the strap on the wearer's skin or clothing, thereby reducing the likelihood of slippage. The adhesive fabric could be a material with a high coefficient of friction, such as rubber or silicone, which would provide a non-slip surface.

[0013] Another proposed modification is to increase the footprint of the arm, leg, or body-worn band on the applicable body part of the wearer. This could be achieved by making the strap wider or designing it to wrap around the arm, leg or body part more times. By increasing the surface area of contact between the strap and the wearer's body, the strap is less likely to slide down, especially when weighted with an attached item.

[0014] These modifications are designed to maintain the position of the attached item and ensure the system's reliability across a range of applications, including carrying Archery Range Finders, Multi-Tools (e.g., "Leatherman Type"), Special Ops tools to clear automatic rifle jams, Special Ops tools for breaching door and window systems, Special Ops Sniper Dope Cards and Windage Indicators, Medical/Surgical Kits for Army/Navy Field Medics, Navigation or Compass Systems, GPS locators, Solar Charging Systems, Auto mechanic tools, Government Employees Badges, and machine gun clearing tools.

[0015] In summary, the disclosed non-slip dual magnet band system is innovatively designed to prioritize magnet-to-magnet attraction, ensuring a robust and reliable connection between personal items and the wearer. This system comprises two key components: a body-worn band and an item-worn band, each embedded with strong magnets. The body-worn band, adaptable for placement on various body parts such as the arm, torso, leg, neck, or head, features an adhesive fabric, clip, pin to fabric, or friction enhancement on its inner surface to prevent slippage, thereby maintaining the item in a fixed position regardless of the wearer's movements. The item-worn band, designed to encircle and secure the intended tool or device, mirrors this magnetic capability, facilitating a silent, quick, and secure attachment through the magnetic force generated between the two bands. This magnetic interaction is central to the system's functionality, allowing for a hands-free carrying experience that is both efficient and effective across a wide range of activities and scenarios, from outdoor adventures to professional tasks requiring quick access to tools or devices.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0016] Other objectives of the disclosure will become apparent to those skilled in the art once the invention has been shown and described. The manner in which these objectives and other desirable characteristics can be obtained is explained in the following description and attached figures in which:

[0017] FIG. 1 is a perspective view of a user **2000** wearing a headband **1010**, a torso band **1020**, an arm band **1030**, a forearm band **1040**, a belt **1050**, and leg band **1060**, where the arm band **1030** has an item **3000** connected to it via dual magnetic force between an item band **1050** and the arm band **1030**;

[0018] FIG. 2 is a perspective view of an example headband **1010**, torso band **1020**, arm band

**1030**, forearm band **1040**, belt **1050**, or leg band **1060**; and,

[0019] FIG. **3** is a perspective view of an example item band **1050**.

[0020] It is to be noted, however, that the appended figures illustrate only typical embodiments of this invention and are therefore not to be considered limiting of its scope, for the invention may admit to other equally effective embodiments that will be appreciated by those reasonably skilled in the relevant arts. Also, figures are not necessarily made to scale but are representative.

#### DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0021] Disclosed is a non-slip dual magnet band system for hands-free carrying of personal items. The system includes two bands, a body-worn band and an item worn band. The body-worn band includes an adhesive fabric, clip, pin to fabric, or friction enhancement to the back of the strap. The body worn band may also include a relatively large footprint on the applicable body part of the wearer. Other details of the disclosed system are described in connection with the figures.

[0022] FIG. **1** is a perspective view of a user **2000** wearing a body-worn band. In some cases the body-worn band could be a headband **1010**, a torso band **1020**, an arm band **1030**, a forearm band **1040**, a belt **1050**, or leg band **1060**. The figure also shows a personal item **300** with an item worn band **1050**. In one use, the arm band **1030** has an item **3000** connected to it via magnetic force between an item band **1050** and the arm band **1030**. The item could also be carried via a magnetic force between any of the body-worn bands depicted in the figure. Suitably, the magnetic force should be between a magnet in the arm band **1030** and another attracted magnet in the item band **1050**.

[0023] FIG. **2** is a perspective view of an example body worn band **1000**. As shown, the band has a body **1001** that carries a magnet or metallic substance **1002**. The internal side has a footprint **1003** and/or friction increasing elements **1004**.

[0024] The body-worn bands **1000** could be constructed from durable materials such as heavy canvas fabric or polypropylene, which may also be waterproof to withstand various environmental conditions. The choice of material should provide a balance between durability, comfort, and flexibility. For instance, neoprene could be used for its elasticity and comfort against the skin, while a nylon blend could offer a lightweight yet strong alternative.

[0025] Within each band, a magnet or magnetic material **1002** is encapsulated to facilitate the magnetic fastening system. The magnets should be strong enough to hold the intended items securely but not so strong as to create excessive magnetic fields that could interfere with electronic devices. Rare earth magnets like neodymium could be used for their strong magnetic properties and compact size.

[0026] Each band should preferably have an upper layer and a lower layer, with the magnetic components (like magnets) disposed between them. The edges of these layers are fastened together to retain the magnetic material securely. The bands could be slip on or over or feature adjustable fastening systems, such as buckles or hook-and-loop systems like Velcro®, to accommodate different user sizes and to ensure a snug fit that prevents slippage. The bands could also incorporate friction-enhancing materials on the inner surface that contacts the user's body or clothing. This could include silicone grip patterns or rubberized textures to prevent the bands from sliding during movement.

[0027] For the headband (**1010**), the design should ensure that it stays in place during vigorous activity without causing discomfort. The torso band (**1020**) or belt (**1050**) should distribute the weight of the attached items evenly to avoid strain. The arm band (**1030**), forearm band (**1040**), or leg band (**1060**) should allow for a full range of motion while keeping the attached items accessible. In summary, the bands described in FIG. **2** would be versatile, durable, and comfortable for the user, with a focus on maintaining a secure attachment of items through an effective magnetic fastening system. The materials and features of each band should be chosen to enhance the functionality and user experience of the fastening system.

[0028] FIG. **3** is a perspective view of an example item band **1050**. As shown, the band has a body

**1051** that carries a magnet or metallic substance **1052**. The internal side has a footprint **1053** and may or may not include friction increasing elements **1004** (not shown in FIG. 3; see FIG. 2).

[0029] The improved non-slip system, which builds upon the dual strip magnetic fastening system described in U.S. Pat. No. 10,786,068 (incorporated by reference as if set forth here in its entirety), could be used to attach a variety of tools and devices across different scenarios. Here are examples of tools that could be attached using the improved system for each scenario: [0030] 1. A Quick Strap System for an Archery Range Finder: The improved strap could be used to attach a compact, lightweight archery range finder, allowing archers to quickly access distance information before taking a shot. [0031] 2. A Quick Strap for a Multi-Tool "Leatherman Type": The system could secure a multi-tool, providing easy access to various tools like pliers, scissors, and screwdrivers for outdoor or survival situations. [0032] 3. A strap for Special Ops to use for a tool to clear automatic rifle jams: A specialized tool for clearing rifle jams, such as a combination of a carbon scraper, bolt carrier scraper, and firing pin cleaner, could be attached for quick maintenance in the field. [0033] 4. A strap for Special Ops to use on legs for breaching door and window systems: Breaching tools like a compact battering ram or a hydraulic door opener could be secured on the leg strap for rapid entry during operations. [0034] 5. A strap for Special Ops Sniper Dope Cards and Windage Indicator: The strap could hold laminated sniper dope cards and a portable windage indicator to assist snipers in making accurate long-distance shots. [0035] 6. A strap for arm use and Medical/Surgical Kits for Army/Navy Field Medics: The strap could be used to attach a medical kit containing essential items such as tourniquets, bandages, and hemostatic agents for immediate field treatment. [0036] 7. A strap for a Compass System: A military-grade lensatic compass could be attached to the strap, providing soldiers with a reliable navigation tool that is always within reach. [0037] 8. A strap for a Location Chip, for tool/equipment locator, and for family to locate individuals who are not home yet: A small GPS tracker could be attached to the strap, allowing for the location of tools, equipment, or even individuals in case of emergency. [0038] 9. A strap for a Phone System & Solar Charging System: A smartphone along with a compact, foldable solar panel charger could be attached, ensuring communication and power availability in remote locations. [0039] 10. A strap for an Auto mechanic that uses only one tool per day: The strap could hold a frequently used tool such as a digital torque wrench, providing convenience and efficiency for the mechanic. [0040] 11. A strap for Government Employees Badge System: A government ID badge or access card could be attached to the strap, allowing for secure and easy access to restricted areas. [0041] 12. A strap with a solar charger and Location chip for a new patent: This strap could integrate a solar charger and a location chip, providing a way to keep electronic devices charged and locatable at all times. [0042] 13. A Chest Strap Double Magnet System for Military Use for items like a machine gun clearing tool: A robust clearing tool for machine guns could be attached to the chest strap, allowing soldiers to quickly address weapon malfunctions during combat. [0043] The improved system's addition of adhesive fabric, clip, pin to fabric, or friction enhancement would ensure that these tools remain securely attached without the strap sliding down, enhancing the functionality and reliability of the system in various operational contexts. [0044] The disclosed non-slip dual magnet band system prioritizes magnet-to-magnet attraction, ensuring a more reliable connection between personal items and the wearer. This system comprises two components: a body-worn band and an item-worn band, each embedded with strong magnets. The body-worn band, adaptable for placement on various body parts such as the arm, leg, waist, torso, or head, features an adhesive fabric, clip, pin to fabric, or friction enhancement on its inner surface to prevent slippage, thereby maintaining the item in a fixed position regardless of the wearer's movements. The item-worn band, designed to encircle and secure the intended tool or device, mirrors this magnetic capability, facilitating a silent, quick, and secure attachment through the magnetic force generated between the two bands. This magnetic interaction is central to the system's functionality, allowing for a hands-free carrying experience that is both efficient and effective across a wide range of activities and scenarios, from outdoor adventures to professional

tasks requiring quick access to tools or devices.

[0045] Although the method and apparatus is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features, aspects, and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead might be applied, alone or in various combinations, to one or more of the other embodiments of the disclosed method and apparatus, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the claimed invention should not be limited by any of the above-described embodiments.

[0046] Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open-ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as meaning “including, without limitation” or the like; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; the terms “a” or “an” should be read as meaning “at least one,” “one or more,” or the like; and adjectives such as “conventional,” “traditional,” “normal,” “standard,” “known,” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that might be available or known now or at any time in the future. Likewise, where this document refers to technologies that would be apparent or known to one of ordinary skill in the art, such technologies encompass those apparent or known to the skilled artisan now or at any time in the future.

[0047] The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to,” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases might be absent. The use of the term “assembly” does not imply that the components or functionality described or claimed as part of the module are all configured in a common package. Indeed, any or all of the various components of a module, whether control logic or other components, might be combined in a single package or separately maintained and might further be distributed across multiple locations.

[0048] Additionally, the various embodiments set forth herein are described in terms of exemplary block diagrams, flow charts, and other illustrations. As will become apparent to one of ordinary skill in the art after reading this document, the illustrated embodiments and their various alternatives might be implemented without confinement to the illustrated examples. For example, block diagrams and their accompanying description should not be construed as mandating a particular architecture or configuration.

[0049] All original claims submitted with this specification are incorporated by reference in their entirety as if fully set forth herein.

## Claims

1. A friction-enhanced, dual strip magnetic fastening system that includes an arm fastener band and a device fastener band, designed to securely attach and retain various tools or devices, such as a hunting accessory.

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